Computer Science (New Book) Class 9th

Important Short Question (Chapter Wise)-2026

Ch # 1:

1. What is an information system?
An Information System is simply an organized set of components that are coordinated to perform a designated function.

2. How system theory can be described?
A branch of a science that deals with complicated structures in living organisms, that relate the human with society and the science is known as Systems Theory.

3. Define natural science.
In natural science, scientists study existing natural systems to understand their workings. **Example:** Natural Science is studying the ecosystem of a forest to understand how different species interact (descriptive).

4. Define design science.
In design science, scientists create new systems (artifacts) to solve problems or achieve specific goals. **Example:** Design Science is developing a new software system to manage forest data and improve conservation efforts (prescriptive).

5. What do you know about single memory store.
Both program instructions and data are stored in the same memory space. **Example:** In a computer game, both the game's code and the data (like scores and player positions) are stored in the same RAM.

6. Define store programming.
Programs are stored in memory and can be changed by the computer. **Example:** When you update a software program, the new instructions replace the old ones in memory.

7. What is the function of CU?
A Control Unit (CU) is a peripheral that governs the activities of the CPU by instructing the ALU and memory to execute tasks according to the program instructions. It ensures the proper and timely execution of duties by all the other components.

8. What is sequential execution?
Instructions are processed one after another in a sequence. **Example:** When your computer runs a program, it follows the steps one by one in the order they are written.

9. Give the examples of a simple system.
A simple example of system is a car, it is made up of an engine, wheels, brakes, and other related items. Every part plays a unique task, but collectively they are responsible for making the car move.

10. Define basic concepts of a system.
Basic concepts will be introduced to emphasize fundamental concepts and principles.

11. What is objective of a system?
Every system has a purpose or goal that it wishes to fulfill. Analyzing a system's operation requires understanding its aim. This insight improves the efficiency and efficacy of the present system.

12. What is objective of transport system?
A transport system aims to transfer people and products securely and effectively between locations.

13. What are the components of a system?
Components are the building blocks of any system. Each component plays a specific role and contributes to the overall functionality of the system.

14. What is environment of a system?
The environment of a system includes everything external to the system that interacts with it. It consists of all external factors that affect the system's operation.

15. What Is Role of communication in a system?
Communication and interaction among system components is key to the functioning of a system. It ensures that components work together in an organized and smooth manner to achieve the system's objectives.

16. Science can be divided into how many types name them.
Science is a systematic way to validate the understanding. Science can be divided into two main types: ★ Natural Science ★ Design Science.

Ch # 2:

1. What is the purpose of a number system?
Numbering systems are essential in computing because they form the basis for representing, storing, and processing data. Different numbering systems help computers perform tasks like calculations, data storage, and data transfer. These systems allow computers to represent various kinds of information, such as text, colors, and memory locations.

2. Convert 83_{10} into binary.
Convert 83 to binary.

83 / 2	=	41	remainder 1
41 / 2	=	20	remainder 1
20 / 2	=	10	remainder 0
10 / 2	=	5	remainder 0
5 / 2	=	2	remainder 1
2 / 2	=	1	remainder 0
1 / 2	=	0	remainder 1

2	83
2	41 – 1
2	20 – 1
2	10 – 0
2	5 – 0
2	2 – 1
2	1 – 0
	0 – 1

The required result in binary is 1010011_2.

3. What is mean by binary number system?
The binary number system is a base-2 number system that consists of digit from 0 and 1. That's why each digit of the number represents a power of 2.

4. What is mean by hexadecimal number system?
The hexadecimal is a base 16 number system with digit number from 0 to 9 and alphabets from A to F; each digit represents 16 to the power of the position of the digit. The letter A to F stand for the numeric value of 10 to 15.

5. What is mean by octal number system?
Octal is a positional numeral system with base eight, which implies that a digit to be used ranges from 0 to 7. The last digit is a single digit power of 8 while the other digits are the coefficients. In the decimal system, the place values starting from the 8^0, 8^1, 8^2 and so on.

6. What is Arithmetic Operation?
Arithmetic operations include addition, subtraction, multiplication and division, and are performed on two numbers at a time. Binary arithmetic operations are similar to decimal operations but follow binary rules. Here is the basic operations:
- Binary Addition
- Binary Subtraction
- Binary Multiplication
- Binary Division

6. Add 1101_2 + 1011_2.

$$\begin{array}{r} 1101 \\ +\ 1011 \\ \hline 11000 \end{array}$$

7. Define ASCII.
ASCII is an acronym that stands for American Standard Code for Information Interchange. It is a character encoding standard adopted for representing in devices such as computers and similar systems that use text. Each alphabet, number or symbol is given a code number between 0 and 127.

8. What is the range of value for an unsigned two bit INTEGER?
The range of values for an unsigned two byte integer is 0 to 65535.

9. Define the Unicode.
Unicode is an attempt at mapping all graphic characters used in any of the world's writing system. Unlike ASCII, which is limited to 7 bits and can represent only 128 characters, Unicode can represent

over a million characters through different forms of encodings such as, UTF-8, UTF-16, and UTF-32. UTF is an acronym that stands for Unicode Transformation Format.

10. Difference between ASCII and Unicode.

ASCII	Unicode
1. ASCII is an acronym that stands for American Standard Code for Information Interchange. **2.** It is a character encoding standard adopted for representing in devices such as computers and similar systems that use text. **3.** Each alphabet, number or symbol is given a code number between 0 and 127.	**1.** Unicode Stands For Universal Character Encoding. **2.** Unicode is an attempt at mapping all graphic characters used in any of the world's writing system. **3.** Unicode can represent over a million characters through different forms of encodings.

11. What is abbreviation of UTF.

UTF is an acronym that stands for Unicode Transformation Format.

12. What is the purpose of text in ENCODING scheme?

Text encoding schemes are essential for representing characters from various languages and symbols in a format that computers can understand and process. Here are some of the most common text encoding schemes used in computers: → ASCII → Extended ASCII → Unicode

13. Divide 1100_2 by 10_2.

```
        1 1 0
 1 0 | 1 1 0 0
      -1 0
      ---------
          1 0
        - 1 0
      ---------
          0 0
```

14. $101_2 * 11_2$.

```
   1 0 1
   x 1 1
  -------
   1 0 1
  1 0 1 x
  -------
  1 1 1 1
```

15. Convert 83 into octal.

8	83
8	10 – 3
8	1 – 2
	0 – 1

The required result in octal which is 123.

16. Define cloud storage.

Cloud Storage Stores files on remote servers accessible via the internet, providing flexibility and backup options.

17. Write the name of storage devices.

These Are → Hard Disk Drive (HDD) → Solid State Drive (SSD) → Cloud Storage

Ch # 3:

1. Define digital system.

Digital systems are the backbone of today's electronics and computing. They manipulate digital information in the form of binary digits, which are either 0 or 1 and are used in calculation devices such as calculators and computers, among others.

2. Define analogue signal.

Analog signals are signals that changes with time smoothly and continuously over time. They can have any value within given range.

Examples: ⮳ Voice signal (speaking) ⮳ Body's temperature ⮳ Radio wave signals

3. Define digital signals.

Digital signals are the signals, which have only two values that are in the form of 0 and 1. These are utilized in digital electronics and computing systems. **Example:** Binary Data In Computers.

4. Difference between analogous signal and digital signals.

Analogue Signal.	Digital Signals
1. Analog signals are signals that changes with time smoothly and continuously over time. **2.** They can have any value within given range.	**1.** Digital signals are the signals, which have only two values that are in the form of 0 and 1.

3. Examples: Voice signal, Body's temperature, Radio wave signals.	**2.** These are utilized in digital electronics and computing systems. **3. Example:** Binary Data In Computers.

5. What is an Analog to Digital conversion?

Analog to Digital Conversion (ADC) is the conversion of analog signals into digital signals, which are discrete and can be easily processed by computerized devices like computers and smart phones.

6. What is digital to analogue conversion?

Digital to Analog Conversion (DAC) is the conversion whereby digital signals are converted to analog signals, making it possible for human to perceive the information, for instance through speakers.

7. Why Is ADC and DAC conversion needed?

Digital to analog conversion, and vice versa, is critical since it enables data processing, storage, and transmission. Digital signals are much less affected by noise and signal degradation and are therefore better suited for transmitting and storing information over long distances. **Example:** Sound Waves

8. Define Boolean Algebra.

Boolean algebra is a branch of mathematics relate to logic and symbolic computation, using two values namely True and False. It is an essential branch of digital circuits since it is the basis for the analysis and design of circuits.

9. What is Boolean function give One example.

Boolean functions are algebraic statements that describe the relationship between binary variables and logical operations. **Example:** Consider a Boolean function with two inputs, A and B. We can construct a function F that represents the AND operation: F (A, B) = A . B

F (A, B) = A . B

10. Define Binary Variables.

Binary variables that can have only have two values, 0 and 1

11. What are Logic Operations?

Logic operations are basic operations implemented in Boolean algebra for processing of these binary variables. The primary logic operations are AND, OR and NOT.

12. Define AND operation.

AND is the basic logical operator which is used in Boolean algebra. It requires two binary inputs which will give a single binary output. The symbol "**.**" is used for the AND operation. The output of the AND operation is "1" only when both inputs are "1" otherwise, the result is "0".

13. Define OR operation.

The OR operation is basic logical operator in Boolean algebra. It requires two binary inputs which will give a single binary output. The symbol "**+**" is used for the OR operation. The output of the OR operation is "1" only when at least 1 of inputs is 1. The output is 0 only when both inputs are '0'.

14. Define NOT operation.

The NOT operation is one of the basic Boolean algebra operations which takes a single binary variable and simply negates its value. If the input is one, the output is zero and if the input is zero, the output is one.

15. Define Truth Table.

A truth table is a table that shows the result of a Boolean expression for all the possible combinations of the values given to the variables related by some logical operators in the expression.

Example: Truth Table for Boolean Function F (A, B) =A.B?

A	B	F (A, B) = A.B
0	0	0
0	1	0
1	0	0
1	1	1

16. Draw the truth table of AND OR NOT operation.

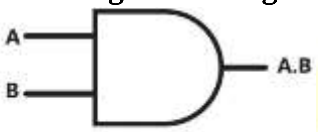
A	B	A AND B (P)
0	0	0
0	1	0
1	0	0
1	1	1

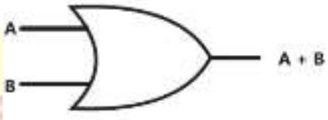
A	B	A OR B (P)
0	0	0
0	1	1
1	0	1
1	1	1

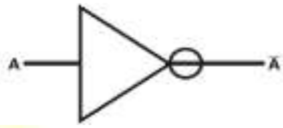
A	NOT A (P)
0	0
0	1

17. Define logic gates.
Logic gates are physical devices in electronic circuits that perform Boolean operations. Each type of logic gate corresponds to a basic Boolean operation. **Examples**: AND Gate, OR Gate, NOT Gate, NAND Gate, etc.

18. Draw the Sign of AND gate, NOT gate, OR gate.

AND 

OR 

NOT 

19. What is the purpose of simplification of functions?
Simplification of Boolean functions is a particularly important process in designing an efficient digital circuit. Such simplified functions require fewer gates making them compact in size, energy efficient and faster than the complicated ones. Simplification means applying of some Boolean algebra rules to make the functions less complicated.

Ch # 4:

1. Define Troubleshooting.
A process of identifying and fixing problems in a system is called system troubleshooting. System troubleshooting is important to ensure smooth operation of any system.

2. What is troubleshooting process?
The troubleshooting process involves several steps that help you systematically identify and fix problems. These steps ensure that you don't overlook any potential issues and that you solve the problem efficiently.

3. What is meant by identification of a problem?
The first step in troubleshooting is to identify the problem. This means recognizing that something is not working as it should. For example, if you press power button and your laptop does not turn on, the problem is clear that it will not start.

4. What is importance of troubleshooting in computer system?
Troubleshooting is very important in computing systems because it helps keep our computers, software, and networks running smoothly. When something goes wrong with a computer system, it can disrupt our work, cause data loss, or even lead to security issues.

5. What are ram failures issues?
Common signs of RAM issues include frequent system crashes, Blue Screens Of Death (BSOD), and poor performance. The computer may also fail to boot or restart randomly

6. What is RAM percentage for failure of computers crashes?
Faulty RAM can cause system crashes and data corruption. RAM errors can account for up to 10% of all computer crashes and Blue Screens Of Death (BSOD).

7. What is data management and data backups?
Data Management and Backups mean storing, and organizing data so it is easy to find and use. It helps make sure the data is available, accurate, and ready when needed.

8. Names of cloud storage services.
Use cloud storage services like Google Drive, Dropbox, or OneDrive to back up your data online. This allows you to access your files from anywhere with an internet connection.

Ch # 5:

1. Define Software.
Software is a collection of programs and instructions that tell a computer what to do and how to do. Without software, computers would be useless machines.

2. Name First Computer Virus And When Was It Created?
The first computer virus, called "Creeper," was created in 1971 as an experimental self-replicating program. It simply displayed the message, "Fm the creeper, catch me if you can!"

3. Write Names Of Types Of Software.
There are Two Types of Software: ★ System Software ★ Application Software
4. Define System Software. Or Define System Software And Provide Two Examples.
System software is designed to manage the system resources and provide a platform for application software to run. It acts as a bridge between the hardware and the user applications.
Here are some Types Of System Software:
- Operating Systems
- Device Drivers
- Utility Programs

5. Write Some Examples Of Operating System.
Examples include Microsoft Windows, macOS, and Linux.
6. Write Some Examples Of Device Drivers.
These include printer drivers, graphics card drivers, and sound card drivers.
7. Write Some Examples Of Utility Programs.
Examples of utility programs are antivirus software, disk cleanup tools, and backup software.
8. Define Application Software. Give Examples.
Application software is designed to help users perform specific tasks. These programs are built to fulfill user needs and are typically more varied than system software.
Examples include:

Word Processors: Such as Microsoft Word and Google Docs.
Web Browsers: Such as Google Chrome, Mozilla Firefox, and Safari.
Games: Such as Minecraft, Fortnite, and Among Us.
Media Players: Such as VLC Media Player and Windows Media Player.

9. Differentiating Between System Software And Application Software With Examples.

System Software	Application Software
Purpose: System software manages and operates computer hardware, making it possible for application software to run.	**Purpose:** Application software helps the user to perform specific tasks.
Examples: System software includes operating systems and device drivers.	**Examples:** Application software includes word processors, web browsers, and games.
Installation: System software is usually pre-installed on a computer	**Installation:** Application software can be installed by the user as needed.

Ch # 6:
1. Define Computer Network.
A computer network is a system of linked devices and computers that may exchange data and operate together.
2. What are the primary objectives of computer network?
The primary objective of computer network is to enable
★ Resource Sharing ★ Data Communication ★ Collaboration
3. Define data communication.
Networks facilitate data transfer, enabling communication through emails, instant messaging, and video conferencing. **Example:** Employees in different locations can collaborate through video conferencing tools like Zoom or Microsoft Teams.
4. What are the role of switch?
Switches play an important role in modern networks by efficiently managing data traffic and ensuring that information reaches the correct device.
5. Write the names of networking devices.
Networking devices include hubs, switches, routers, and access points, which are responsible for the management and direction of network traffic.

6. Define Router.

A router is a networking device that interconnects networks or allows devices to connect to it. It directs data packets between different networks. It is a traffic director on the internet, making sure that data gets from one place to another efficiently.

7. Define DHCP.

DHCP automatically assigns IP addresses to devices on a network, simplifying network management. **Example:** When a device connects to a Wi-Fi network, DHCP assigns it an IP address.

8. Define TCP IP.

TCP/IP (Transmission Control Protocol/Internet Protocol) is the fundamental suite of protocols for internet communication.

9. Write the names of data communication system components.

It comprises of five basic components:

→ Sender → Receiver → Message → Protocol → Medium

10. What is Access Point (AP)?

An Access Point (AP) is a networking device that facilitates the connection of wireless devices to a wired network. It works as a link between your computers and smartphones or any other wireless Router device and the internet.

11. Define Nodes.

Devices that are connected to the network, such as computers, smartphones, and printers.

12. Define links.

The connections between nodes, which can be wired (like Ethernet cables) or wireless (like Wi-Fi).

13. What is resource sharing?

Computer networks allow devices to share resources, such as printers and storage, reducing costs and improving efficiency. **Example:** In an office network, multiple computers can share a single printer, reducing the need for multiple printers.

14. Define protocol.

Protocols are sets of rules that govern data communication. Common protocols include TCP/IP, HTTP, FTP and SMTP. **Example:** HyperText Transfer Protocol (HTTP) is used for transferring web pages over the internet.

15. Define topology.

Network topologies are methods used to define the arrangement of different devices in a computer network, where each device is called a node. The reliability and performance of a network are impacted by the way its devices are linked.

16. What Are The Advantages of Star Topologies?

In a star topology:

1. Each node in network communicates with the other central switch or hub.
2. It is easy to install.
3. In star topology if one node is down, it doesn't affect whole network.
4. The hub works as a data flow repeater In Star Topology.

17. Which top topology provides high redundancy and reliability?

In a Mesh topology, each device is connected to every other device. This provides high redundancy and reliability. **Example:** Imagine a city where every house is directly connected to every other house by roads. If one road is blocked, there are multiple alternative routes.

18. What Is Transmission Modes? Write types.

Network communication modes describe how data is transmitted between devices. There are three primary modes: ⚐ Simplex ⚐ Half-Duplex ⚐ Full-Duplex

19. Define OSI Model. Write Name Of layers.

The Open Systems Interconnection (OSI) Model is a framework used to understand how different networking protocols interact. It has 7 layers, each with a specific function. Name of each layers are:

1. Physical Layer

2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

20. Define LAN, MAN, CAN.

LAN: A LAN is a network that connects computers and devices within a limited area, such as a home, school, or office building. **Example:** The computer network in your school that connects all the computers in the lab is a LAN.

MAN: A MAN is a network that spans a city or a large campus, connecting multiple LANs together. **Example:** The network that connects various branches of a university across a city is a MAN.

CAN: A CAN is a network that connects multiple LANs within a limited geographical area, such as a university campus or a business park. **Example:** The network that connects various departments and buildings within a university is a CAN.

Ch # 7:

1. Define Computational Thinking.

Computational Thinking (CT) is a problem-solving process that involves a set of skills and techniques to solve complex problems in a way that a can be executed by a computer.

2. What is decomposition in competition thinking?

Decomposition is the method of breaking down a complicated problem into smaller, more convenient components. Decomposition is an important step in computational thinking. It involves dividing a complex problem into smaller, manageable tasks.

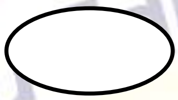
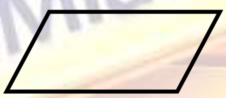
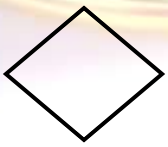
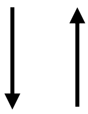
3. What is an algorithm?

An algorithm is a step-by-step collection of instructions to solve a problem or complete a task. **Or** An algorithm is a precise sequence of instructions that can be followed to achieve a specific goal.

4. Define flowchart.

Flowcharts are visual representations of the steps in a process or system, depicted using different symbols connected by arrows.

5. Draw and name any two symbols of flowcharts.

Symbol	**Name**
	Oval (Terminals)
	Rectangle (Process)
	Parallelogram (Input/Output)
	Diamond (Decision)
	Arrow (Flow Line)

6. Define Pseudo Code.

Pseudocode is not actual code that can be run on a computer, but rather a way to describe the steps of an algorithm in a manner that is easy to follow. It helps programmers and students focus on the logic of the algorithm without worrying about the syntax of a specific programming language.

7. Write any two difference between pseudo code and flowchart.

Sr.	Pseudocode	Flowcharts
1.	Pseudocode uses plain language and structured format to describe the steps of an algorithm.	Flowcharts use graphical symbols and arrows to represent the flow of an algorithm.
2.	It is read like a story, with each step is written out sequentially.	It is like watching a movie, where each symbol (such as rectangles, diamonds, and ovals) represents a different type of action or decision, and arrows indicate the connection and direction of the flow.
3.	Pseudocode communicates the steps in a detailed, narrative-like format.	Flowchart communicates the process in a visual format, which can be more intuitive for understanding the overall flow and structure.
4.	It is particularly useful for documenting algorithms in a way that can be easily converted into actual code in any programming language.	They are useful for identifying the steps and decisions in an algorithm at a glance.

8. What is Dry Run?

A dry run involves manually going through the algorithm with sample data to identify any errors.

It Used As: → Dry Run Of A Flowchart → Dry Run Of Pseudocode

9. Define Simulation.

Simulation is we use of computer programs to create a model of a real-world process or system. This helps us understand how things work by testing different ideas or algorithms without needing to try them out in real life.

10. What are LARP stands for?

LARP stands for Logic of Algorithms for resolution of Problems.

11. How many types of errors?

There are three main types of errors:

- Syntax Errors
- Runtime Errors
- Logical Errors

12. Define Syntax error.

These occur when we write something incorrectly in our algorithm or flowchart. For example, missing a step or using the wrong symbol.

13. List the names of DEBUGGING techniques.

Some common debugging techniques are:

1. Trace the Steps 2. Use Comments 3. Check Conditions 4. Simplify the Problem

14. Define debugging.

Debugging is the process of finding and fixing errors in an algorithm or flowchart.

15. What is pattern recognition and give an example?

Pattern recognition involves looking for similarities or patterns among and within problems. **Example:** If you notice that you always forget your homework on Mondays, you might recognize a pattern and set a reminder specifically for Sundays.

16. Define abstraction and its importance in problem solving.

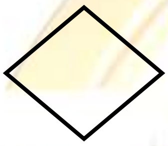
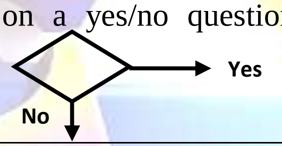
Abstraction is the process of hiding the complex details while exposing only the necessary parts.

Importance: It helps reduce complexity by allowing us to focus on the high-level overview without getting lost in the details. This helps in understanding, designing, and solving problems more efficiently.

17. What is Problem Simplification?
Simplifying a problem involves breaking it down into smaller, more manageable sub-problems. **Example:** To design a website, break down the tasks into designing the layout, creating content, and coding the functionality.
18. What is clarity in importance of flowchart?
Clarity: Flowcharts provide a clear and concise way to represent processes, making them easier to understand at a glance.
19. What is the purpose of parallelogram in flowchart symbols?

	Parallelogram (Input/Output)	A parallelogram represents data input or output (e.g., reading input from a user or displaying output on screen. **Example:** 

20. What is the purpose of diamond symbols in flowchart?

	Diamond (Decision)	A diamond shape represents a decision point in the process where the flow can branch based on a yes/no question or true/false condition. **Example:** 

21. What is time complexity in algorithm evaluating?
Time complexity is usually expressed using Big O notation, like O(n), O(logn), or $O(n^2)$. It helps us compare different algorithms to see which one is faster!
22. What is the Space complexity in algorithm evaluating?
Space complexity measures the amount of memory an algorithm uses relative to input size. It is essential to consider both the memory required for the input and any extra memory used by the algorithm.
23. Write the names of algorithm evaluating techniques.
Designing and evaluating algorithms involves activities like dry runs and simulations to ensure they work as intended.
24. Why is LARP Important?
LARP helps you:
1. Understand how algorithms work.
2. See the effect of different inputs on the output.
3. Practice writing and improving your own algorithms.

25. Define Runtime error.
Runtime error happens when the algorithm or flowchart is being executed. Example: Trying to perform an impossible operation, such as dividing by zero.
Ch # 8:
1. What is web development?
Process of creating websites and web applications is called Web development. It means using various programming languages and tools to design, build, and maintain websites.
2. Define Web Servers.
Web Servers are computers that store and deliver web pages to users when they enter a URL
3. Define Databases.
Databases store and manage data, like user information, product details, and website content.
4. Why HTML used.
HTML is the standard language used to create web pages. Think of HTML as the building blocks of a website. Just like LEGO pieces (as shown in Figure) come together to build a structure, HTML tags come together to build a web page.
5. Why text editor used in HTML.
To start creating websites, you need a few basic tools and environments:

Text Editor: This is where you write your HTML code. Popular text editors include Notepad+ +, Sublime Text, and Visual Studio Code.

6. Name any two text editors name for HTML.

Popular text editors include Notepad+ +, Sublime Text, and Visual Studio Code.

7. What are web browsers?

You will use this to view and test your HTML files. Common web browsers are Google Chrome, Mozilla Firefox, and Microsoft Edge.

8. Define HTML tags.

Elements that make up an HTML document are called tags. A web page's structure and content are defined by them. On the basis of structure, HTML Tags are categorized into two types:

9. How many tags of HTML?

Paired tags: Comes in pairs an opening Tag and closing Tag i.e <p> .....</p>.

Unpaired Tags: Do not need closing Tags. They are also known as self-closing Tags. i.e., <img>,
.

10. What is the range of heading in HTML?

Headings in HTML, ranging from <h1> to <h6>, are used to define the structure and hierarchy of content on a web page.

11. Why links are used in HTML?

Links in HTML are used to connect one web page to another. They allow you to click on words or images to go to different parts of the same web page or to other web pages on the internet.

12. What is javascript?

JavaScript is a programming language that is used to make websites interactive and engaging. It allows developers to create things like animations, games, and responsive features that react when you click buttons or move your mouse.

13. What is the purpose of function in Java script?

Functions allow you to reuse code and perform specific tasks. They are like mini-programs that you can run whenever you need them.

Example: An example of a simple function that displays a greeting message:

```
        <Script>
        function greet(){
        alert("Hello, Student!");
        }
        greet(); //This calls the function to execute
</script>
```

14. What is the purpose of head tag in HTML?

Every HTML document has a basic structure where: **<head>:** This section contains meta-information about the HTML document, like the title.

15. What is the difference between an order list and an unordered list in HTML?

Ordered List	Unordered List
1. An ordered list keeps each list item with an order number. **2.** If you change the order, the meaning of the whole list may also change. **3. Tag**: An ordered list starts with ** ** and ends with ** ** tag. Each list item starts with ** ** tag.	**1.** In an unordered list, the order of the list items is not important. **2.** In other words, shuffling of items in an unordered list has no effect. **3. Tag**: An unordered list is created inside the ** ** tags. Each list item is added with ** ** tag.

Example:

HTML code	Output
** ** ** - First Item ** ** - Second Item ** ** - Third Item ** ** **	1. First Item 2. Second Item 3. Third Item

Example:

HTML code	Output (Result)
** ** ** - Item 1 ** ** - Item 2 ** ** - Item 3 ** ** **	• Item 1 • Item 2 • Item 3

16. What is the function of div tag in HTML?

HTML elements like <div> and <section> are used to group content together. You can then use CSS to style and position them. **Example:** <div class = "container">

<section class = "header" > This is the header</section >
<section class = "content" > This is the main content</section >
<secti0n class = "footer" > This is the footer</section >
</div>

17. Which tag is used insert table in HTML?

A table is created using the <table> tag, which contains rows defined by <tr> (table row) tags, and each row consists of cells represented by <td> (table data) tags. Additionally, headings for the table can be added using <th> (table header) tags to provide context for the data.

18. What is Search Engine Optimization SEO?

Search engines use headings to understand the structure and main topics of a web page. Proper use of headings can improve the page's SEO, helping it rank higher in search results.

19. What is a benefit of comments in HTML?

In HTM L, comments can be extremely useful for:

★ Explaining the purpose of a specific section of code
★ Leaving reminders for future edits
★ Temporarily disabling code for testing purposes

20. CSS stands for.

CSS Stands for Cascading Style Sheets.

21. Why CSS is important?

1. CSS is very important for improving the visual appearance of webpages.
2. It improves user experience.
3. CSS allows web developers to control the colors, fonts, and layout.
4. CSS offers various properties and selectors to apply styles to specific elements.
5. CSS enabling responsive design that automatically adjusts to different screen sizes and devices.

22. Describe the syntax of creating a hyperlink in HTML.

Links are created using the <a> tag.
<a href= "https://www. example.com" > Visit Example.com< /a>
<a href= " mailto : example @ example.com">Send Email

Ch # 9:

1. Define Data.

Data consists of raw facts collected about things around us that we can process to generate useful information. It can take many forms, such as numbers, words, measurements, observations, or even images and sounds, and may originate from various sources.

2. How many types of data can be divided?

Data can be divided into two broad categories namely:

- Qualitative Data
- Quantitative Data

3. What is Qualitative Data?

Qualitative data refer to categories or labels used to describe the qualities or characteristics of something rather than its quantity. This type of data offer a way to categorize and provide insights into opinions, behaviors, and experiences through descriptions rather than numbers.

4. What is Quantitative Data?

Quantitative data consists of numbers used to measure the quantity or amount of something. These data types answer questions like "How much?" or "How long?" and can be useful for mathematical calculations and statistical analyses.

5. What is the difference between qualitative and quadratic data?

Qualitative Data	Quantitative Data
1. Qualitative data refer to categories or labels. **2.** It is used to describe the qualities or characteristics of something rather than its quantity. **3.** This type of data offer a way to categorize and provide insights into opinions, behaviors, and experiences through descriptions rather than numbers.	**1.** Quantitative data consists of numbers. **2.** It is used to measure the quantity or amount of something. **3.** These data types answer questions like "How much?" or "How long?" and can be useful for mathematical calculations and statistical analyses.

6. Define Charts.

Charts are visuals representation of data designed to make complex information easier to understand. Charts help identify patterns, trends and outliers in datasets.

7. Write some common types of charts.

Common types of charts are:

→ Bar charts → Line charts → Pie charts

8. Define Graphs.

Graphs are visuals tools used to represent data and show relationship between different data points.

9. What are common types of graphs?

Common types of graphs are:

→ Line graphs → Bar graphs → Scatter plots → Histograms

10. Define Data Collection.

Data collection is the process of gathering information to answer questions, make decisions, or understand something better. There are different methods and tools for collecting data, each with its own way of gathering and recording information**.** The methods of data collection are:

✓ Surveys ✓ Questionnaires ✓ Interviews
✓Observations ✓ Online Data Sources

11. Define Survey.

Surveys collect information from people by asking them questions. This can be done on paper, over the phone, or online. **Example:** To find out your classmates favorite ice cream flavors, you might create a survey with questions like "What is your favorite ice cream flavor?" and give it to your classmates to fill out.

12. What is the purpose of interviews?

Talking to individuals one-on-one to gather detailed information. **Example:** Interviewing a school teacher to understand their experience and challenges.

13. What is Observation?

Watching and noting what happens in a particular situation. **Example:** Observing how students behave during a group project to understand how they work in a team.

14. What type of data is the number of students in your class?

The number of students in your class is discrete data because it involves countable, distinct values.

15. Define Data Visualization.
Data visualization is the process of turning numbers and information into pictures. These pictures make it easier for us to understand what the data is telling us. When we look at data in the form of charts or graphs, it becomes simpler to see patterns, trends, and relationships.
16. What is Data Backup?
Data Backups are copies of important data or files stored separately from the original to protect against data loss. Backups are essential to ensure that you can recover your data if something goes wrong, such as accidental deletion, hardware failure, or a computer virus.
Ch # 10:
1. Define Artificial Intelligence.
Artificial Intelligence is a rapidly growing field that is transforming various aspects of our lives. From healthcare to gaming, AI technologies are being applied to solve complex problems and improve our daily experiences. **Example:** AI-driven systems monitor crop health and predict yields by getting data from sensors and drones to optimize farming practices.
2. Define IoT.
IoT is a revolutionary concept that is transforming the way we live and work. It involves connecting everyday devices and systems to the internet, allowing them to communicate and interact with each other.
3. Provide two examples of AI application in healthcare.
AI is used for diagnosing diseases, personalizing treatment plans, and predicting patient outcomes.
4. How AI applications used in education?
AI-powered tools provide personalized learning experiences, automate administrative tasks, and offer insights into student performance. AI tools assist in teaching students, checking their homework or assignments and tracking performance.
5. How AI used in Social Media?
AI is extensively used in social media to power personalized content recommendations, automated content generation, sentiment analysis, user behavior prediction, and targeted advertising to enhance user engagement and optimize marketing strategies.
6. How AI Integrated in E-commerce?
AI is highly integrated into e-commerce platforms, powering personalized product recommendations, intelligent chatbots for customer support, fraud detection systems, and others.
7. What is Machine Learning?
Machine learning is a type of artificial intelligence where computers learn from experience and improve over time without being explicitly programmed. It's like teaching a computer by showing it lots of examples, and it figures out how to do things on its own.
8. What is Deep Learning?
Deep learning is a special kind of machine learning. It uses complex structures called neural networks, which are inspired by how our brains work. These networks help computers learn from lots of data and make decisions or recognize patterns even better.
9. Define Robotics.
Robotics is the science of building and programming robots. Robots are machines that can do tasks for us, like cleaning the floor or building cars. Some robots can even think and make decisions.
10. How Many Categories of AI Algorithms?
Artificial Intelligence (AI) involves using algorithms and techniques to enable machines to perform tasks that typically require human intelligence. AI algorithms can be broadly categorized into two types based on their interpretability:

1. Explainable (Whitebox) Algorithms **2.** Unexplainable (blackbox) Algorithms

11. What is the Significance of IoT?
IoT is significant because it allows for the seamless integration of the physical and digital worlds. This connection enables devices to collect and share data, which can be analyzed to improve

efficiency, provide better services, and create new opportunities in various fields such as healthcare, agriculture, and smart homes.

12. Give examples of IoT used in transportation.

IoT is enhancing transportation systems, making them more efficient and safer. Connected vehicles, smart traffic lights, and real-time tracking systems are some examples of how IoT is used in transportation.

13. Describe the significance of IoT in connecting devices and systems.

IoT is significant because it allows for the seamless integration of the physical and digital worlds. This connection enables devices to collect and share data, which can be analyzed to improve efficiency, provide better services, and create new opportunities in various fields such as healthcare, agriculture, and smart homes.

14. Role of AI applications in gaming.

AI enhances game design, creates realistic characters, and improves player experiences.

15. What is the difference between Machine Learning and Deep Learning?

Machine Learning	Deep Learning
1. Machine learning is a type of artificial intelligence where computers learn from experience and improve over time without being explicitly programmed. **2.** It's like teaching a computer by showing it lots of examples, and it figures out how to do things on its own.	**1.** Deep learning is a special kind of machine learning. It uses complex structures called neural networks, which are inspired by how our brains work. **2.** These networks help computers learn from lots of data and make decisions or recognize patterns even better.

16. What is Natural Language Processing NLP?

Natural Language Processing, or NLP, is a technology that helps computers understand and talk to us in our language. It's like teaching a computer to read, write, and even chat with us.

Example: When you ask Siri or Alexa a question, they use NLP to understand what you're saying and give you a helpful answer. Another example is when you type a message and your phone suggests words to complete your sentence. That's NLP at work!

17. How many components of IoT Systems?

An IoT system typically consists of the following components:

- Sensors
- Actuators
- Devices
- Networks
- Data Analysis

18. Define Sensors.

These are devices that detect and measure physical properties like temperature, humidity, light, and motion. Sensors collect data from the environment.

19. What is Actuators?

These are devices that convert energy into motion-lm IoT an actuator can act on data to generate output.

20. What is Data Analysis?

This involves processing and analyzing the data collected by sensors to gain insights and make decisions. Data analysis can be done on the device itself, in the cloud, or on a central server.

Ch # 11:

1. What Are The Use of Computer In Daily Life?

Computers have become a crucial part of our everyday lives. Whether we are using them for schoolwork, chatting with friends, or playing games.

2. What are Computer Hardware?

When we talk about computers, hardware refers to the physical parts like the monitor, keyboard, and CPU.

3. Give two example of Software.

Software includes the programs and applications we use, such as word processors or games.

4. What does mean for Secure Use of Digital Platforms?
Using digital platforms securely means taking extra steps to protect your information and ensuring that your online activities do not put you or others at risk.
5. What is two Factor Authentication?
Two-Factor Authentication adds an extra layer of security to your accounts. After entering your password, you will be asked to provide another piece of information, like a code sent to your phone. This makes it much harder for someone to hack into your account.
6. What is Baking Up Important Data?
Regularly backing up your data ensures that you will not lose important information if something goes wrong with your device. You can back up your data to an external hard drive or a cloud storage service like Google Drive or Dropbox.
7. What is Online Scams?
Online scams are designed to trick you into giving away your personal information. These can include phishing emails that pretend to be from legitimate companies asking for your login details. Always be skeptical of unsolicited requests for personal information, and verify the source before responding.
8. What is the Role of Social Media?
Social media platforms like Facebook, Instagram, and Twitter allow us to connect with friends and share information. However, it is important to think before you post. Always avoid sharing personal information, like your home address or phone number, publicly.
9. Define Patent.
Patents protect new inventions or processes, giving the inventor exclusive rights to make, use, or sell the invention for a certain period. **Example:** If someone invents a new type of smartphone, they can patent it to prevent others from making or selling a similar phone without permission.
10. What are Cloud Services?
Cloud services like Google Drive or Dropbox allow you to store and share files online. These services are convenient, it is important to use them wisely. Always use strong passwords to protect your accounts and avoid sharing sensitive information, like passwords or financial details, through cloud storage.
11. Define Copyright.
Copyright is a legal right that gives creators control over their original works, such as music, books, movies, and software. **Example:** When an author writes a book, they have the copyright to decide how the book is published, shared, or adapted. This means no one else can copy or distribute the book without the author's permission.
12. Why Intellectual Property Rights Are Important?
Intellectual property rights are important because they protect the creations and ideas of individuals and organizations. When someone creates something new, like a piece of music, a book, or an invention, they have the right to control how it is used. Some Intellectual property rights are:

- Copyright
- Trademarks
- Patents

13. Give two Examples of Social Media.
Social media platforms like Facebook, Instagram, and Twitter allow us to connect with friends and share information.
14. Why should you Create Strong Password for your Account?
To save and protects you from harm and avoids any unwanted issues. Always create strong, unique passwords for your accounts. A strong password typically includes a mix of letters, numbers, and special characters.
Example: Instead of using "password123," you could use something like "B3tterP@sswOrd!".
15. What is the purpose of regular software updates?
Keeping your devices and applications up to date is crucial for safety. Software updates often include important security fixes that protect your device from new threats.

16. Why we avoid using public Wi-Fi for Sensitive Transactions?
Public Wi-Fi networks, like those in cafes or libraries, are often less secure. It is best to avoid accessing sensitive information, such as online banking, while connected to these networks. Instead, wait until you're on a secure, private network at home.
17. What is Trademark?
Trademarks are symbols, names, or slogans used by companies to distinguish their products or services from others. **Example:** The Nike "swoosh" logo is a trademark. Trademarks protect brand identity, so no other company can use a similar symbol to confuse customers.
Ch # 12:
1. What is the meaning of Word Entrepreneurship?
The word "entrepreneur" comes from a French word that means, "to undertake". Entrepreneurs undertake the task of starting and running new businesses.
2. What do you mean by Innovation?
Innovation means creating something new or improving something that already exists. Entrepreneurs are always looking for new ways to solve problems or make things better.
3. Why is Entrepreneurship Important?
Entrepreneurship is important because it drives economic growth, creates jobs, and fosters innovation. New businesses bring fresh ideas and competition, which can lead to better products and services for everyone.
4. Write some Social Media Platforms. Name them.
Social media platforms such as Facebook, Instagram, and Twitter allow entrepreneurs to market their products, engage with customers, and build a brand presence.
5. What is Cloud Computing?
Cloud computing allows businesses to store data and run applications over the Internet, reducing the need for physical infrastructure and enabling remote work.
6. What is Market Analysis?
Market analysis involves researching your target market to understand the needs and preferences of your potential customers. This includes studying market trends, analyzing competitors, and identifying your target audience.
7. What is Ethical Entrepreneurship?
Ethical entrepreneurship involves incorporating principles of ethics into all aspects of business operations and decision-making. This includes honesty, integrity, fairness, and respect for people and the environment.
8. What is an E-Commerce platforms?
E-commerce platforms enable businesses to sell products and services online. They provide tools for setting up online stores, managing payments, and offering customer support.
9. What is Importance of E-commerce in Business?
E-commerce platforms enable businesses to sell products and services online. They provide tools for setting up online stores, managing payments, and offering customer support.
10. What are Revenue Models?
A revenue model outlines how your business will generate income. This includes pricing strategies, sales forecasts, and potential revenue streams.
11. What are Tech Startups?
Tech startups like Facebook, Google, and Apple began as small companies founded by entrepreneurs who had innovative ideas for new technology. These companies have grown to become some of the largest and most influential in the world.
12. Give two examples of entrepreneurship?
Google, Apple, Amazon, Draz, Bykea.

13. What is the role of digital technology in entrepreneurship?
Digital technologies provide entrepreneurs with tools and platforms that can enhance their business operations, reach a global audience, and improve efficiency. Some of them Are:
✓ Social media ✓ Mobile Apps ✓ Cloud Computing
✓ Big Data Analytics ✓ Digital Marketing And E-commerce

14. What are Sustainable Development Goals SDGs?
The Sustainable Development Goals (SDGs) are a set of 17 global goals established by the United Nations to address various social, economic, and environmental challenges. Businesses can play a crucial role in achieving these goals by aligning their strategies with sustainability.

15. What is Social Sustainability?
Ensuring fair labor practices, promoting education, and supporting community development.

16. What is Environmental Sustainability?
Reducing carbon footprints, using renewable resources, and minimizing waste.

17. What is Economic Sustainability?
Creating jobs, fostering innovation, and contributing to economic growth.

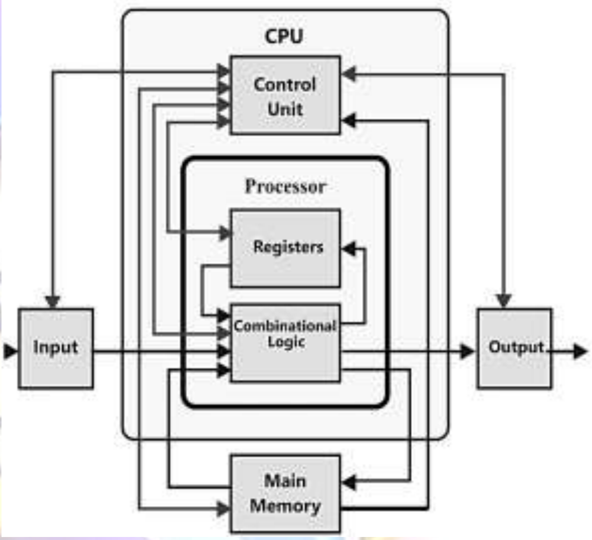
Important Long Question (Chapter Wise)-2026

Chapter # 1:

LQ # 1: What Is The Von Neumann Computer Architecture? List Its Key components. OR
Write A Note On Computer Primary Components In Term of Von Neumann. OR
Explain The Picture Of Von Neumann Architecture.

The Von Neumann architecture is a computer paradigm that delineates a system in which the hardware of the computer has four primary components:

★ The Memory
★ The Central Processing Unit (CPU)
★ Input Mechanisms
★ Output Mechanisms.



This model is called the John von Neumann model, the Neumann model named in honor of the mathematician and physicist who contributed to its development during the 1940s.

Components: The key parts that constitute the architecture of the Von Neumann computer are as:

➢ Memory
➢ Central Processing Unit (CPU)
➢ Input Devices
➢ Output Devices

1. Memory: Memory Contains both input data and the instructions (program) required for CPU processing. For instance, consider the RAM of your computer, when a program starts it is loaded into RAM to enable faster execution compared to when it runs from the hard disk.

2. Central Processing Unit (CPU): Central Processing Unit (CPU) Performs addition and subtraction, and executes commands provided by the memory. The system has two main components:

- The Arithmetic Logic Unit (ALU)
- The Control Unit (CU)

The Arithmetic Logic Unit (ALU): The Arithmetic Logic Unit (ALU) performs mathematical computations and logical operations.

Control Unit (CU): A Control Unit (CU) is a peripheral that governs the activities of the CPU by instructing the ALU and memory to execute tasks according to the program instructions. It ensures the proper and timely execution of duties by all the other components.

Working Of ALU And CU Together: When doing the calculation 2 + 2 on a calculator application, the Arithmetic Logic Unit (ALU) handles the numerical values while the control Unit (CU) supervises the whole procedure.

3. Input Devices: Input devices enable the users to input data and instructions into the computer system. e.g. Keyboard, Mouse, and Microphone. Entering text from the keyboard transmits data to the CPU for subsequent processing.

4. Output Devices: Output devices present or communicate the outcomes of the tasks executed by the computer. e.g. Monitor, Printer. Consider, for instance, a monitor and printer. Upon completion of data processing, the CPU transmits the outcome to the monitor for visual display.

<u>LQ # 2:</u> Explain The Working Of Von Neumann Architecture. OR
Explain The Von Neumann Fetching, Decoding, Execution, Storing.

The Von Neumann architecture encompasses four essential stages for a CPU to carry out instructions, namely retrieval, interpretation, execution and storage.

✓ Fetching ✓ Decoding ✓ Execution ✓ Storing

1. Fetching: The central processing unit retrieves an instruction from the computer's memory. This instruction specifies the operation to be executed by the CPU.

Hardware Components: Memory, CPU (Program Counter PC), Instruction Register (IR).

Specification: The Program Counter (PC) stores the memory address of the subsequent instruction. Once the address is stored in memory, the instruction located at that location is retrieved and placed into the Instruction Register (IR).

2. Decoding: In order to determine the necessary action, the Control Unit (CU) decodes the instruction.

Comprising Components: Control Unit (CU)

Detail: The control unit (CU) decodes the opcode (operation code) of the instruction and determines the required procedures and data.

3. Execution: The CPU processes the instruction. When the instruction involves a computation, it is executed by the Arithmetic Logic Unit (ALU). Any task that requires transferring data between several locations is managed by the CU.

Involved Components: ALU, CU.

Detail: The Arithmetic and Logic Unit (ALU) carries out mathematical and logical calculations, while the Control Unit (CU) handles data transmission activities.

4. Storing: The outcome of the computation is either returned to memory or sent to an output device.

Involved Components: Memory and Output Device.

Specification: The outcome is either stored in a designated memory location or sent to an output device, such as a display.

<u>LQ # 3:</u> Write Down The Advantages and Disadvantages of Von Neumann Computer Architecture.

<u>Advantages:</u>

The Advantages of Von Neumann computer architecture are discussed here.

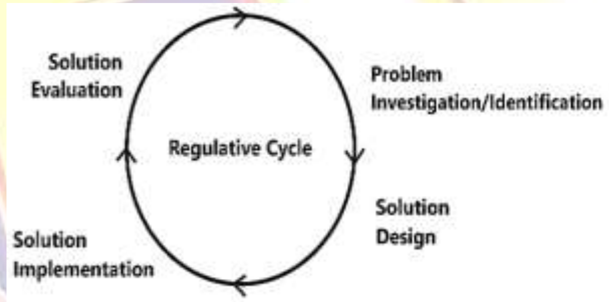
1. **Simplified Design**
   By combining instructions and data into a single memory area, architecture is simplified.
2. **Flexibility**
   Programs can be easily changed by changing memory contents.

Disadvantages:

The Disadvantages of Von Neumann computer architecture are:

1. **The Von Neumann Bottleneck:** The Von Neumann bottleneck occurs when a single memory area limits the CPU's ability to retrieve instructions and data quickly.
2. **Security Risks:** Having data and instructions stored in the same area poses a problem where one program can alter another's instructions in a manner that is security risk.

LQ # 4: Write Down The Difference Between Natural Science And Design Science.

Natural Science	Design Science
1. Scientists study existing natural systems to understand their workings.	**1.** Scientists create new systems (artifacts) to solve problems or achieve specific goals.
2. Natural science is meant to uncover the objectivity and functionality of natural systems in the natural world.	**2.** Design Science is focused on designing and creating artifacts (tools, systems, methods) to achieve specific goals.
3. Its nature is descriptive, meaning that the scientists seek to understand and describe natural phenomena.	**3.** The nature of design science is prescriptive, meaning that it aims to prescribe and create artificial systems.
4. To achieve this, natural scientists follow the empirical cycle of natural science, as shown in Figure. 	**4.** To achieve this design science researchers follow the regulative cycle. 
5. Natural Science is studying the ecosystem of a forest to understand how different species interact (descriptive).	**5.** Design Science is developing a new software system to manage forest data and improve conservation efforts (prescriptive).

Chapter # 6:

LQ # 1: What Are The Primary Components Of Computer Network? OR
Write Down The Basic Components Of Computer Networks.

Computer Network:
A computer network is a system of linked devices and computers that may exchange data and operate together.

Components of Network:
The primary components of network include:

♠ Nodes ♠ Links ♠ Switches ♠ Routers

Nodes:
Devices that are connected to the network, such as computers, smartphones, and printers.

Links:
The connections between nodes, which can be wired (like Ethernet cables) or wireless (like Wi-Fi).

Switches:
Switches are the devices that connect multiple nodes within a network to forward data.

Routers:
Routers are the devices that connect different networks and direct data packets between them.

LQ # 2: What Are The Fundamental Concepts in Data Communication? OR
Write Down The Basic Concept Of Data Communication.

Data Communication:
Data communication involves the exchange of data between a sender and a receiver through a communication medium.

Basic Components Of Data Communication:
It comprises of five basic components:

→ Sender → Receiver → Message → Protocol → Medium

Sender:
The device that sends the data. **Example:** A computer sending an email.

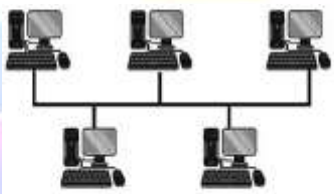
Receiver:
The device that receives the data. **Example:** A smartphone receiving the email.
Message:
The data being communicated. **Example:** The content of the email.
Protocol:
A set of rules governing data communication. **Example:** The HTTP protocol used for web communications.
Medium:
The physical or wireless path through which data travels. **Example:** Ethernet cable or WiFi.

LQ # 3: What Are Network Topologies? Write Its types. OR Define Topology. Explain Any Three Of Its Types.

Network Topologies:
Network topologies are methods used to define the arrangement of different devices in a computer network, where each device is called a node. The reliability and performance of a network are impacted by the way its devices are linked.

Types Of Network Topologies?
There are Following Types Of Network Topologies:

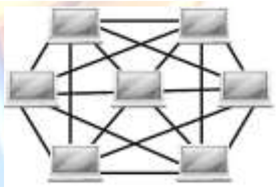
- ✓ Bus Topology
- ✓ Star Topology
- ✓ Ring Topology
- ✓ Mesh T5opology



Bus Topology:
In a Bus topology, all devices share a single communication line called a bus. Each device is connected to this central cable.
Example: Imagine a chalkboard in a classroom where every student can see the notes written by the teacher. Here, the chalkboard represents the shared communication line.

Star Topology:
In a star topology each node in network communicates with the other central switch or hub. The hub works as a data flow repeater.
Example: Think of a school principal's office connected to all classrooms through intercoms. The principal's office is the hub, and the classrooms are the nodes.

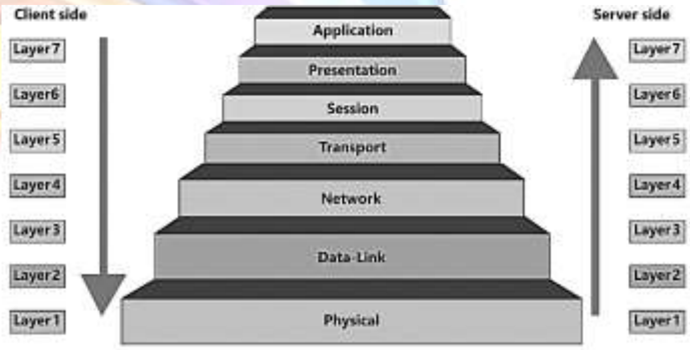
Ring Topology:
In a Ring topology, each device is connected in a circular pathway with other devices. Data travels in one direction, passing through each device.
Example: Consider a relay race where each runner passes the baton to the next runner in a circle until it reaches the starting point again.



Mesh Topology:
In a Mesh topology, each device is connected to every other device. This provides high redundancy and reliability.
Example: Imagine a city where every house is directly connected to every other house by roads. If one road is blocked, there are multiple alternative routes.

LQ # 4: Define OSI Networking Model? Explain Its Layers.

The Open Systems Interconnection (OSI) Model is a framework used to understand how different networking protocols interact. It has 7 layers, each with a specific function.



Name Of 7 Layers Of OSI Networking Model.

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

Physical Layer (Layer 1):
The Physical Layer is liable for the actual connection between devices. The process of sending unprocessed data bits via a physical medium is the focus here.
Example: Imagine the hardware that connects computers, like a Network interface cables, repeaters, hubs and connectors.
Data Link Layer (Layer 2):
Error detection and correction, as well as node-to-node data transport, are handled by the Data Link Layer. It ensures error-free data transmission from the Physical Layer.
Example: Think of the Data Link Layer as traffic lights at intersections, which manage the flow of cars (data) and prevent collisions.
Network Layer (Layer 3):
The Network Layer is responsible for data transfer between different networks. It determines the best path for data to travel from the source to the destination.
Example: Imagine a GPS system finding the best route for you to travel from home to school.
Transport Layer (Layer 4):
The Transport Layer ensures that data is transferred from one process rerunning on source end system to a process souring on destination end system. It manages data flow control and error checking. The Transport Layer uses protocols like Transmission Control Protocol (TCP) to ensure reliable data transfer!
Example: Think of the Transport Layer as a delivery service that ensures your package arrives safely and on time.
Session Layer (Layer 5):
The Session Layer manages sessions between applications. It establishes, maintains, and terminates connections between devices.
Example: Imagine a phone call where the session layer sets up the call, keeps it connected, and ends it when you hang up.
Presentation Layer (Layer 6):
The Presentation Layer translates data between the application layer and the network. It formats and encrypts data to ensure it is readable by the receiving system. The Presentation Layer handles data encryption and compression!
Example: Think of the Presentation Layer as a translator converting a book from one language to another so that more people can read it.
Application Layer (Layer 7):
The Application Layer is the closest to the end user. It provides network services directly to applications, such as email, web browsing, and file transfer.
Example: Imagine the Application Layer as a waiter taking your order in a restaurant and bringing your food.

LQ # 5: What Are Transmission Modes? Write Details. OR
Describe The Transmission Modes.

Network communication modes describe how data is transmitted between devices.
Types Of Transmission Modes:
There are three primary modes:
- Simplex Communication
- Half-Duplex Communication
- Full-Duplex Communication

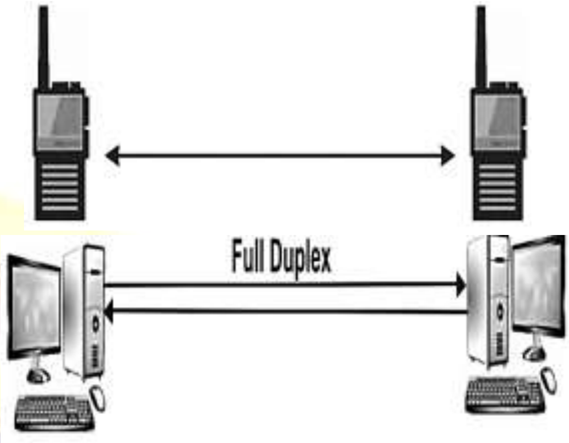
1. Simplex Communication:
In Simplex communication, data transmission is unidirectional, meaning it flows in only one direction. A advice can either send or review data in this communication.

Example: Keyboard to computer is an example of simplex communication.

2. Half-Duplex Communication.

In Half-Duplex communication, data transmission can occur in both directions, but not simultaneously. One device must wait for the other to finish transmitting before it can start.

Example: Walkie-talkies



3. Full-Duplex Communication:

Full-duplex communication allows for simultaneous data delivery in both directions. Both devices may transmit and receive data simultaneously at the same time.

Example: Telephone conversations are an example of Full-Duplex communication. Both people can talk and listen at the same time without waiting for their turn.

<u>LQ # 6:</u> What Are Internet Protocol (IP) Addresses? Write Its Versions? OR Write The Difference Between Ipv4 And Ipv6.

Internet Protocol (IP) addresses are unique identifiers assigned to devices connected to the Internet. There are two primary versions:

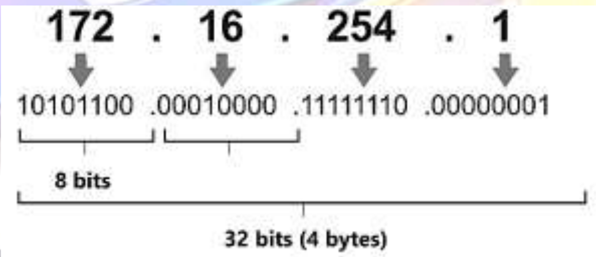
- Internet Protocol Version 4 (IPv4)
- Internet Protocol Version 6 (IPv6)

Internet Protocol Version 4 (IPv4):

IPv4 is the fourth version of the Internet Protocol and the most widely used today. It uses a 32-bit address scheme, allowing for approximately 4.3 billion unique addresses. ‘

Total Number Of Unique IPv4 Addresses:

To find the total number of unique IPv4 addresses, we calculate 2^{32}, which represents all possible combinations of 32 bits, i.e. $2^{32} = 4,294,967,296$.



Internet Protocol Version 6 (IPv6):

IPv6 is the most recent version of the Internet Protocol designed to replace IPv4. It uses a 128-bit address scheme, allowing for an almost limitless number of unique addresses.

Example: Imagine an IPv6 address like a digital fingerprint. It can provide a unique identifier not just for houses on a street, but also for every grain of sand on a beach. e.g. 2001:0000:130F:0000:0000:0900: 876A:130B.

Best of Luck (Board Exams 2026)

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